Journal of the Geological Society, London, Vol. **157,** 2000, pp. 317–325. Printed in Great Britain.

Palaeomagnetic constraints on the age of the Red Arch Formation and associated sandstone dykes (Northern Ireland)

P. TURNER[1], R. SHELTON[2], A. RUFFELL[2] & J. PUGH[1]
[1]*School of Earth Sciences, The University of Birmingham, Birmingham B15 2TT, UK*
[2]*Department of Geology, School of Geosciences, The Queens University of Belfast, Belfast BT7 INN, UK*

Abstract: The Red Arch Formation is a continental red bed succession of uncertain age exposed on the northeast coast of Ireland. There are lithological similarities with the Old Red Sandstone of the Highland Border region, although unconformities within the succession, the presence of reworked aeolian sand grains and variations in the induration of the rocks have prompted earlier workers to suggest that the upper part of the succession may be Triassic in age.

This paper reports palaeomagnetic results from the Red Arch Formation. The characteristic magnetizations (ChRMs) are exclusively of reversed polarity and the palaeomagnetic pole position for the sequence bears close comparison with other results from the Old Red Sandstone in Scotland and southern Britain. These poles are regarded as magnetic overprints formed during the Kiaman Superchron because they fall in the Permo-Carboniferous segment of the apparent polar wander path.

Sandstone dykes that cut the Red Arch Formation on a NNE–SSW and NNW–SSE trend have also been studied palaeomagnetically. The dykes show more steeply inclined ChRMs predominantly of normal polarity. The dykes are therefore interpreted to be post-Kiaman in age, a conclusion consistent with our structural observations. The age of the dykes cannot be determined precisely because they have also been, at least partially, remagnetized. Lithological and regional geological considerations suggest the dykes are Triassic in age and formed in an extensional stress regime prior to the opening of the North Atlantic. They were remagnetized in Tertiary times, possibly associated with uplift and emplacement of the British Tertiary Igneous Province.

Keywords: Old Red Sandstone, Kiaman, Antrim, paleomagnetism.

An outstanding problem in the geological analysis of the sedimentary fill to the basins in Northern Ireland is the distinction between two red bed successions of Devonian and Permian–Triassic age. This problem is especially acute near major faults where non-marine or shallow marine oxidized clastic successions of the two sequences may be juxtaposed. Although the Devonian and Permian–Triassic successions can contain biostratigraphicalIy significant microfossils in Northern Ireland and adjacent areas, these tend to occur at discrete horizons and at only a few locations (see references below). The bulk of either succession remains largely undated and mapped correlations depend on tectonic location and lithofacies.

There are a number of reasons why dating these successions is important. In assessing the burial history of the Carboniferous coals and oil shales that form the major potential hydrocarbon source rock in the area, knowing whether to include significant thicknesses of sediment that may overlie the source (Permian–Triassic) or underlie it (Old Red Sandstone) is critical. In addition, previous workers (Simon 1984*a*, *b*; Kerr 1987; McCaffrey *et al*. 1993) have used fracture orientations in the red bed successions to assess the structural controls on deposition and the kinematics of later deformation; hence determining the age of the fractured rock is important. It is the structure, palaeomagnetic properties and age of the putative Upper Palaeozoic (ORS) and Permo-Triassic red beds flanking the extension of the Highland Boundary Fault Zone in Northern Ireland which are the focus of this paper.

Regional stratigraphy and context of the Red Arch Formation

The Precambrian and Lower Palaeozoic rocks of the Caledonides in northern Ireland are covered by, or faulted against, substantial thicknesses (kilometers) of supposed Devonian and younger sedimentary rocks which were deposited in the western continuation of the Midland Valley of Scotland. These sedimentary successions include the Old Red Sandstone facies of the Devonian, and Carboniferous clastics, carbonates and organic-rich shales and coals. The Permian–Triassic saw the development of new sedimentary basins that formed in response to post-Variscan rifting (Jackson & Mulholland 1993). The Permian–Triassic fill which dominates these basins is mainly composed of non-marine or shallow-marine, evaporite-bearing clastic rocks with minor lavas and dolomitic limestones. Minor inversion episodes separate these and the overlying thin (tens to hundreds of metres thick) Cretaceous Ulster White Limestone (Chalk equivalent) and the flood basalts of the Antrim Lava Group. The youngest sedimentary succession is areally restricted and relates to a separate period of subsidence centered on Lough Neagh (the Oligocene Lough Neagh Group clays and lignites).

The best exposed succession of juxtaposed (presumed) Devonian and later Permian–Triassic sediments in Northern Ireland occurs between Cushendun and Waterfoot in Country Antrim. Conglomeratic beds of the Red Arch Formation are exposed between Cushendall and Waterfoot (Fig. 1) where Simon (1984*b*) interpreted the succession as the product of

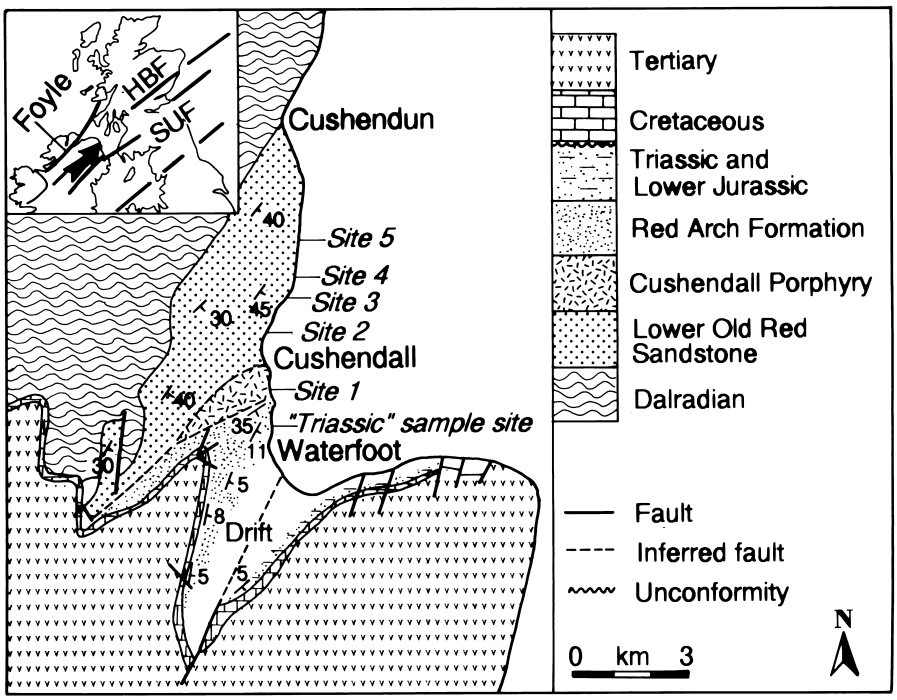


Fig. 1. Regional location and geological map of the study area, redrawn partly after Simon (1984*b*).

a small, complex alluvial fan, building in a southeasterly direction off the Highland Border Ridge. The active fault would therefore have been the continuation of the Highland Boundary Fault Zone into Ireland. Sanderson (1970) has documented that the Highland Border Ridge remained largely emergent from the Devonian onwards, and should therefore be considered to be a sediment source area both at this time and later.

Sedimentary rocks deemed to be Lower Old Red Sandstone in age (Cross Slieve Group) are exposed to the north of Cushendall towards Cushendun (Fig. 1). Simon (1984*a*) discussed the provenance of these strata, although no direct evidence was presented to verify the Lower Old Red Sandstone age. A porphyry obscures the contact between the Cross Slieve Group and the Red Arch Formation. This paper is concerned only with the Red Arch Formation.

An informal lithostratigraphy for the Red Arch Formation has been established by Simon (1984*b*) and comprises five units (Fig. 2) within the context of an overall fining-upwards succession. Unit 1 consists of an alternation of conglomerates and pebbly sandstones with the former containing clasts up to 1 m in diameter. The contact between units 1 and 2 is unconformable (Unconformity A) and Unit 2 consists mainly of massive clast-supported conglomerates. The contact between Units 2 and 3 is not exposed. Unit 3 consists of moderately sorted, pebbly sandstones, interfingering with clast-supported conglomerates. Unit 4 reverts back to thick clast-supported conglomerates, and finally Unit 5 comprises matrix-supported conglomerates and sandstones. The contact between Units 4 and 5 is also considered to be an unconformity (Unconformity B, Simon 1984*b*), but it could also be interpreted as a fault. Clasts throughout the succession include schists, quartzites and reworked conglomerate.

Previous interpretations of the age of the strata

Simon (1984*b*) interprets the section between Cushendun and Waterfoot to be entirely of Upper Old Red Sandstone age based on a lack of marine influence and comparison with similar sections in western Scotland. However, a number of workers (e.g. Symes & McHenry 1886; Wilson 1953; Kerr 1987) have suggested that the the strata exposed above Unconformity A belong to the Triassic (Fig. 2) as the beds here are softer and contain abundant reworked aeolian quartz grains. However, the hardness phenomenon may simply be a localized diagenetic artefact, and it is clear that reworked aeolian grains are not restricted to the Triassic. Kerr (1987) interpreted that the strata above Unconformity A as Triassic because: (a) these beds have a shallower dip than the underlying beds and (b) the orientation of sandstone dykes within the formation fit into the known stress system for the Triassic of Britain, corresponding to an ENE–WSW to E–W extension (Ziegler 1982). However, the change in structural dip could reflect synsedimentary tectonics (and hence an intrabasinal unconformity developed entirely within an Old Red Sandstone succession), and the dykes may not be syndepositional.

Mitchell & Turner (1996) have recently suggested that the strata could even belong to Lower Carboniferous. In support of this argument they cite two points; firstly in the Fintona Block to the west of Lough Neagh, analagous conglomerates have proved to be Lower Carboniferous on the basis of palynological evidence (Mitchell & Owens 1990). Secondly, no published evidence exists for the succession in NE Antrim being of Devonian age.

Sandstone dykes

A number of sandstone dykes are exposed in the coastal section at Cushendall (a typical example is shown in Fig. 3b). The dykes are generally thin with a thickness ranging from a few millimetres to a few centimetres, with a modal thickness between 0.5 and 1.0 cm. They are generally normal or near normal to bedding and can be traced laterally for distances up to several metres before tapering out. The orientation of the dykes and associated fractures is shown in Fig. 4. They have a

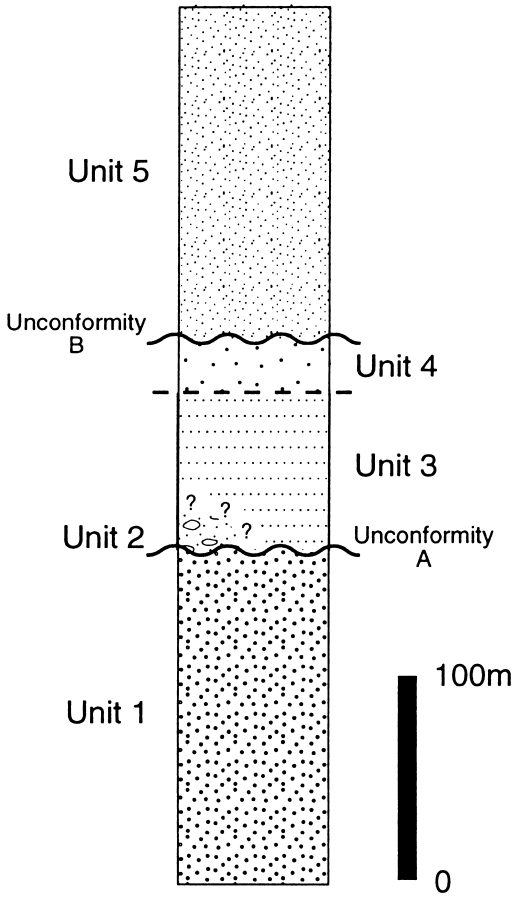


Fig. 2. Stratigraphic section of the Red Arch Formation exposed between Cushendall and Waterfoot, N. Ireland. Redrawn after Simon (1984*b*).

predominantly NNW or NNE strike and show some evidence of variation above and below Unconformity A in that the NNW-trending dykes are more abundant below the unconformity (Fig. 4a). However, Simon (1984*b*) has observed one dyke cut time, this unconformity and suggested they were all emplaced at the same time. He interpreted the dykes to be synsedimentary in origin and hence of Old Red Sandstone age. This is in contrast to the interpretation of Kerr (1987) who favoured a Triassic synsedimentary origin for the dykes as they fit into the regional Triassic stress field, i.e. ENE to E–W extension. Kerr (1987) also described fractures in the Red Arch Formation with a similar orientations to the sandstone dykes (Fig. 4b), although there are also E–W-striking fractures which the dykes have not exploited.

The sandstone dykes and surrounding sediments were subject to petrographical analyses in order to gain a better understanding of their relationships. The results show important differences between the dykes and the host matrix. The host sediment is a poorly sorted arkosic quartz arenite with subangular clasts. It is well cemented with early iron oxide, quartz overgrowth and clay cements with some dolomite blocking pore throats. There is about 5% remaining porosity. The sandstone dykes are mineralogically more mature. Whilst there is a small amount of feldspar, the composition is dominated by quartz and lithic sand grains set in a fine dolomite cement; sometimes the dolomite is displacive, enclosing floating quartz grains; quartz overgrowths are absent in the sandstone dykes.



Fig. 3. Photograph of a sandstone dyke indicating that the matrix was lithified prior to dyke emplacement. Note that the dyke cuts a fractured Dalradian clast and must have been emplaced after lithification and fracturing.

The early nature of the dolomite cement in the sandstone dykes suggests it may be synchronous with the extensional event that allowed intrusion. The absence of fault reversals or strike-slip offsets suggests that the dykes were emplaced during pure extension. The dykes are interpreted as having being emplaced when the host conglomerate was already lithified, rather than synsedimentary emplacement into a soft host. Two lines of evidence support this interpretation. Firstly the intrusion of a dyke across the cleavage in a clast of Dalradian schist (Fig. 3b) suggests that the matrix to the conglomerate was lithified prior to fracturing and dyke intrusion. Secondly a number of fracture orientations exist within the formation (Fig. 4b). Comparison with the dyke orientations (Fig. 4a) shows a coincident fracture population oriented NNW–SSE. The dykes appear to have exloited an existing brittle fracture set. A similar fracture orientation was observed in sediments presumed to be Devonian in age to the north of Cushendall by Kerr (1987) (Fig. 4c).

Palaeomagnetic analysis

Sampling and results

Sampling locations are shown on Fig. 5. A total of 16 orientated hand samples were collected including sedimentary dykes. A summary table

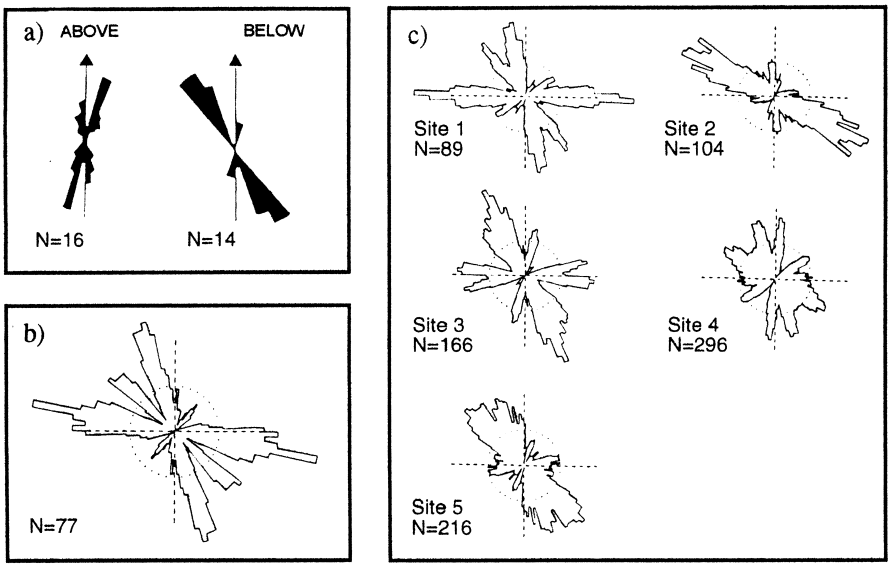


Fig. 4. (**a**) Rose diagrams for the sandstone dykes above and below unconformity A, showing a NNE–SSW and NNW–SSE trend. (**b**) Rose diagram for fractures within the 'Triassic' part of the Cushendall section of Kerr (1987). (**c**) Rose diagrams for fractures within the 'Lower Devonian' to the north of Cushendall (from Kerr 1987).

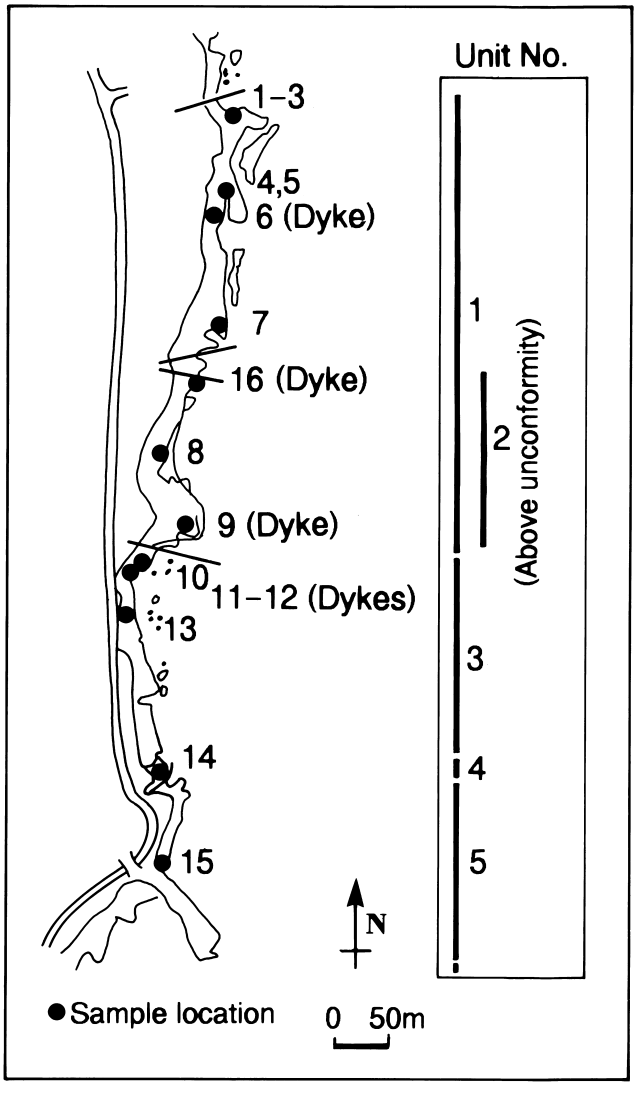


Fig. 5. Location of palaeomagnetic sites and stratigraphic unit analysed.

showing the stratigraphic location of the samples, the sample type and palaeomagnetic components identified is shown in Table 1. From the hand samples a total of 29 standard specimens (2.54 × 2.20 cm) were prepared for analysis. They were measured using a Molspin fluxgate magnetometer and subjected to partial thermal demagnetization using a Magnetic Measurements microprocessor-controlled demagnetizer. All the specimens were heated in 50°C steps up to 650°C and remeasured after each step. The initial susceptibility was also measured at each temperature increment in order to monitor any magnetomineralogical changes which might have taken place in the heating process. The specimens showed initial NRM intensities of 0.3–15 mAm^{-1} and initial susceptibilities between 20 and 200 SI units.

There is little difference between the dykes and the host rock in terms of the modal intensities (1.5–2.0 Am^{-1}) but one of the dykes (CA16) has much stronger remanence than the other samples. The distribution of the initial NRM directions is shown in Fig. 6. They show a poorly defined great circle distribution along a NE–SW axis with a concentration of

Table 1. *Sample details and palaeomagnetic components*

Sample no.	Unit	Type	No. specimens	Components
CA 1	1	SST	2	2B
CA 2	1	SST	1	B
CA 3	1	SST+dyke	1	A+B
CA 4	1	SST	2	2B
CA 5	1	SST	1	B
CA 7	1	SST+dyke	1	A+B
CA 6	1	Dyke	3	3A
CA 8	2	SST	2	2B
CA 9	2	Dyke	2	2A
CA 10	3	SST	1	B
CA 11	3	Dyke	2	2A
CA 12	3	Dyke	3	2A
CA 13	3	SST	2	A+2B
CA 14	4	SST	1	B
CA 15	5	SST	1	B
CA 16	2	Dyke	1	2A

Isolated components are designated A (low unblocking temperature) or B (high unblocking temperature). The geographic location is shown in Fig. 5.

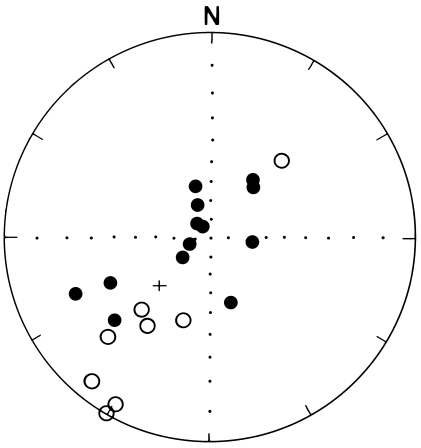


Fig. 6. Stereographic projection showing the distribution of initial NRM directions for all palaeomagnetic specimens (solid symbols are downward dip; open symbols are upward dip). Note that the directions describe a broad great circle distribution.

directions in the SW quadrant with moderate negative (upward) inclination and in the northern hemisphere with steep positive downward inclination. During partial thermal demagnetization the specimens show a variety of different types of behaviour but there are distinct differences between the sandstone dykes and the host rocks.

An example of the host rock (without a sandstone dyke) is shown in (Fig. 7; sample CA2). It shows a direction in the SW quadrant with moderately steep negative (upward) inclination. There is a steady, almost linear loss of NRM intensity between 150°C and 450°C. Only one component of magnetization can be isolated using principle components analysis and this has a mean direction of Dec: 213°; Inc: $-16°$. Other samples from the host rocks show the presence of two components of magnetization. An example of this type is shown in Fig. 8 (CA 13A). In this case the original direction has a positive inclination and there is a relative increase in the normalized intensity up to 450°C. Above this temperature the intensity falls rapidly to 650°C. The orthogonal vector plot shows two components which are separated by a sharp break in their unblocking temperature spectra. The isolated components correspond to Dec: 62°; Inc: $+62°$ and Dec: 241°; Inc $-35°$ although the latter is poorly defined with a mean angular deviation (MAD) of 21°. These two components comprise the characteristic remanence (ChRM) of the host rocks and are designated the A component and the B component respectively.

The relative proportions of the A and B components varies according to how much sandstone dyke material there is in the specimen. In some samples which consist exclusively of sandstone dyke, only the A-component is present. In the example shown in Fig. 9, the ChRM comprises a single component which lies very near to the present-day field direction. More

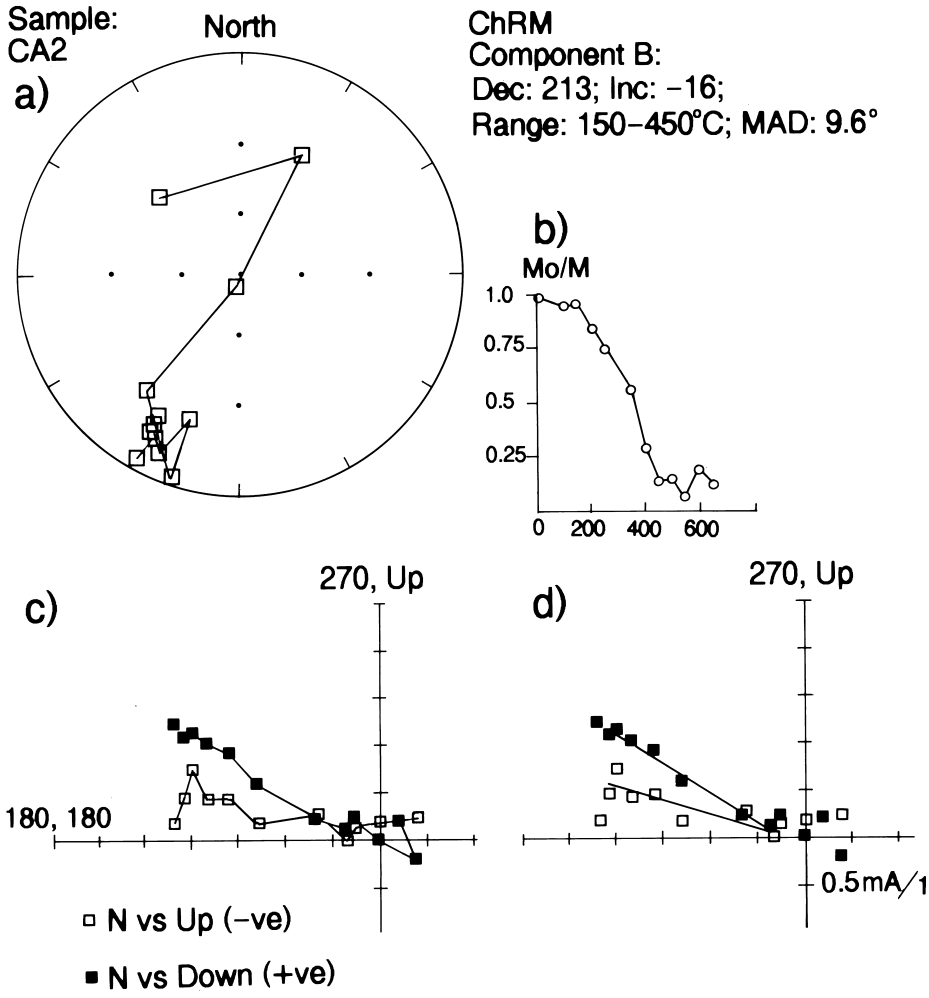


Fig. 7. Palaeomagnetic results for specimen CA2: (**a**) stereographic projection of the directional changes during partial thermal demagnetization (solid symbols are downward dip; open symbols are upward dip; (**b**) normalized intensity decay curve; (**c**) orthogonal vector plot (units are 0.5 mA m^{-1}); (**d**) orthogonal vector plot with direction isolated using principle components analysis.

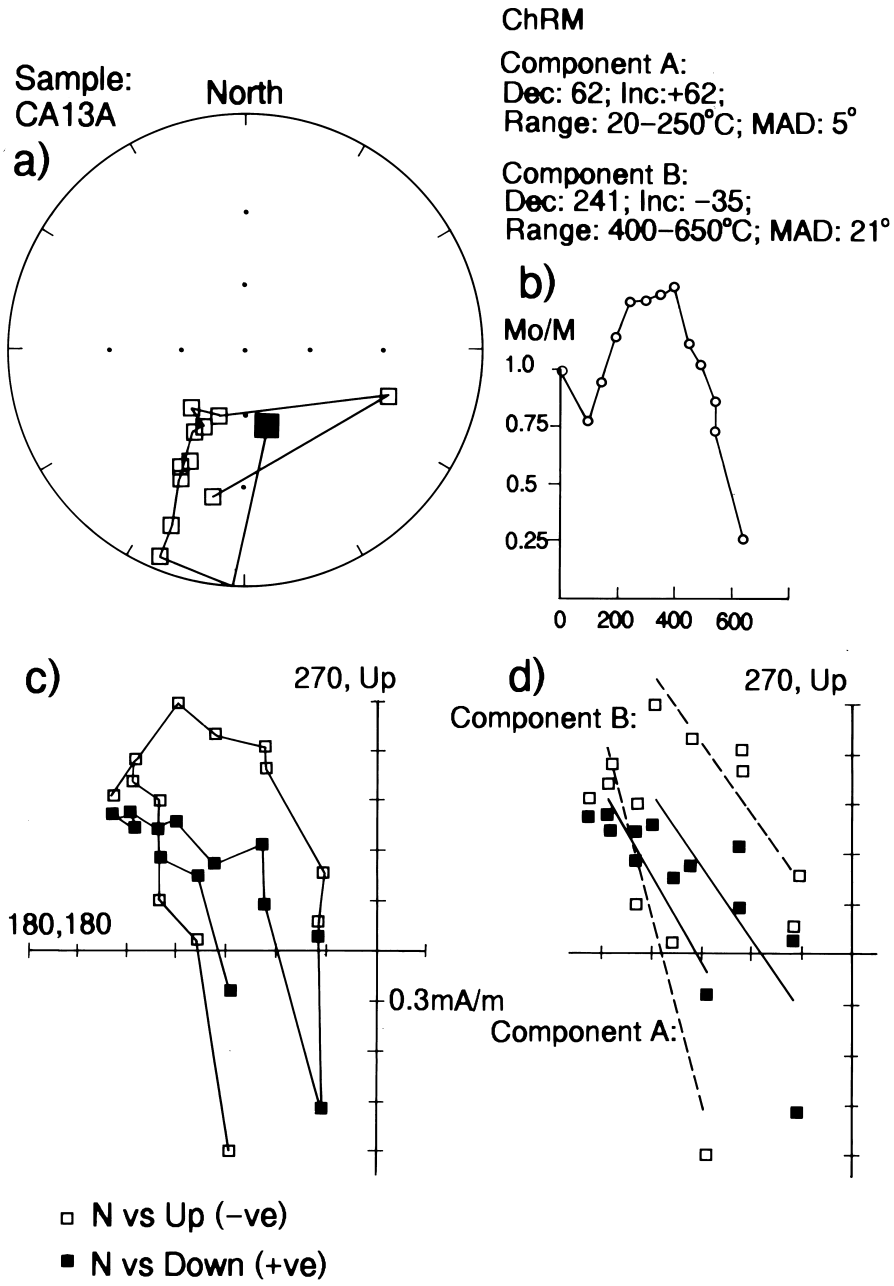


Fig. 8. Palaeomagnetic results for specimen CA13A: (**a**) stereographic projection of the directional changes during partial thermal demagnetization (solid symbols are downward dip; open symbols are upward dip; (**b**) normalized intensity decay curve; (**c**) orthogonal vector plot (units are 0.3 mA m^{-1}); (**d**) orthogonal vector plot with direction isolated using principle components analysis.

than 50% of the remanence remains below 550°C but above this temperature there is a rapid loss of remanence.

In general the A component is characterized by a lower unblocking temperature spectrum and is exclusively of normal polarity with a relatively steep inclination. The directions cluster close to the present day field direction for the area. The B component has a higher unblocking temperature spectrum and is exclusively of reversed polarity with moderately shallow inclination (Fig. 10). Pole positions calculated for the two components correspond to lat: 83°N, long: 346°E (A component, $n=14$, dp,dm=16/18) and lat: 44° S, long: 313°E (B component, $n=15$, dp,dm=7/13).

Palaeomagnetic interpretation

The age of A and B components were estimated by comparison with reference poles from other studies from the British area. The pole position calculated for the B component compares closely with many of the poles reported for continental red beds in the Old Red Sandstone of the British area (Turner & Archer 1975; Turner 1980; Channell *et al.* 1992; Table 2 and Fig. 11). These poles can be seen to compare most closely with those of late Carboniferous–early Permian age and indicate widespread remagnetization of the Old Red Sandstone during this time. This remagnetization appears to be absent from the sandstone dykes, a feature which is consistent with a post-Permian age for the origin of the dykes. The pole position is consistent with a Tertiary–Recent age and compares with those described from the Tertiary igneous province of Scotland (Harris 1983).

Discussion

It is suggested that the Red Arch Formation is of an undifferentiated Old Red Sandstone age, in line with the interpretation

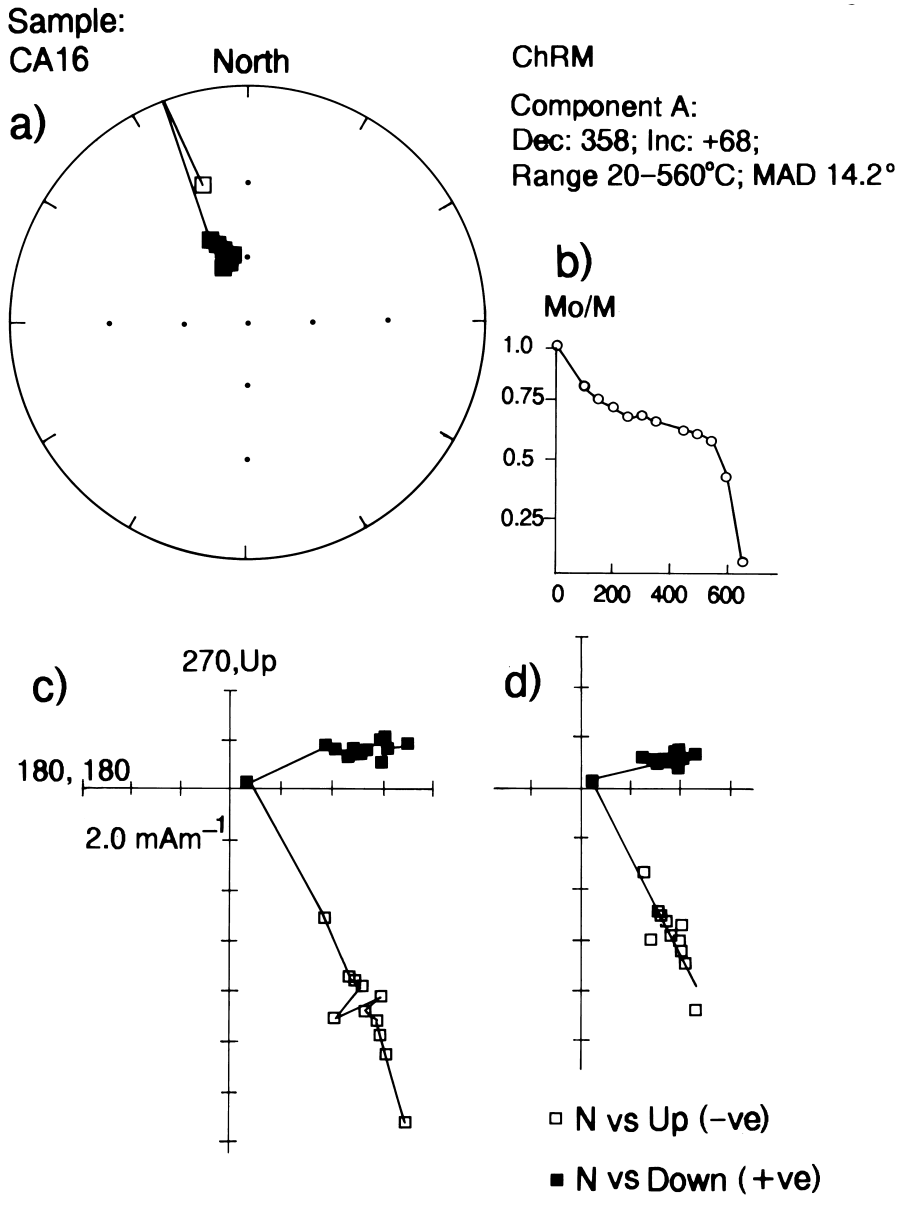


Fig. 9. Palaeomagnetic results for specimen CA16: (**a**) stereographic projection of the directional changes during partial thermal demagnetization (solid symbols are downward dip; open symbols are upward dip; (**b**) normalized intensity decay curve; (**c**) orthogonal vector plot (units are 2.0 mA m^{-1}); (**d**) orthogonal vector plot with direction isolated using principle components analysis.

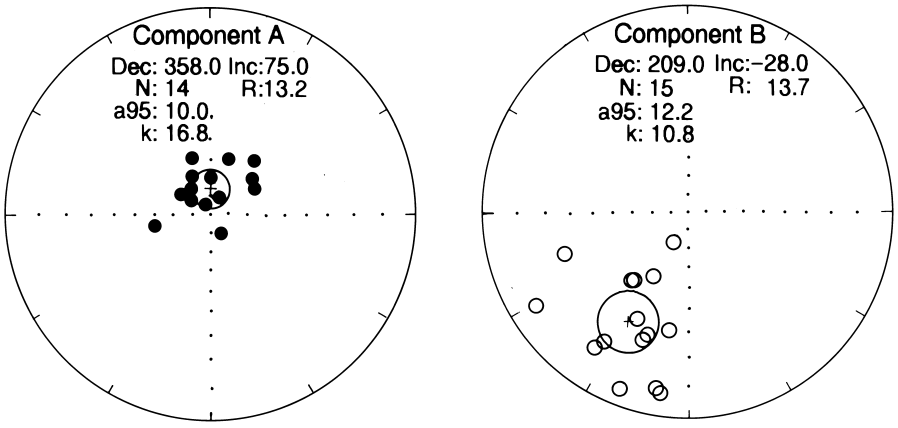


Fig. 10. Stereographic projections showing the distribution of isolated components A and B from the studied specimens.

of Simon (1984*b*). The rocks were remagnetized in Lower Permian times in a similar fashion to that seen in many other Old Red Sandstone basins. The clastic dykes are interpreted to be of post-Permian age as they were not subjected to remagnetization in the Lower Permian. In fact palaeomagnetic components isolated from the sandstone dykes are most

Table 2. *Palaeomagnetic poles used in the interpretation*

No.	Location	Lat	Long	dp/dm	Reference
1	Exeter Lavas	50	329	2/4.3	Zijderveld (1967)
2	Peterhead dyke	41	342	0.7/1.3	Torsvik (1985)
3	Whin Sill	44	339	Not given	Storetvedt & Gidskehaug, (1969)
4	South Wales	40	328	3.8/7.5	Channel *et al.* (1992)
5	Gamrie	45	315	10/17	Turner & Archer (1975)
6	Bristol area	32	338	5/10	Morris *et al.* (1973)
7	Cushendun	44	313	7/13	Component B, this study

Poles 1–3 are late Carboniferous/early Permian igneous rocks and 4–6 are known overprints of the Old Red Sandstone (see Fig. 11).

closely comparable with a Tertiary age. However the sandstone dykes themselves may have been overprinted by a Tertiary or more recent magnetization and a Late Permian or Triassic age cannot be ruled out. In fact the overlying cover of the Cretaceous Ulster White Limestone and Tertiary Antrim Lava Group suggests that a Triassic age is more likely for the dykes.

With regard to the mechanism of emplacement of the sandstone dykes the following sequence of events is suggested on the basis of the field and palaeomagnetic evidence: (1) lithification of Old Red Sandstone conglomerates followed by (2) the creation of a pervasive fracture system cutting through the matrix and clasts and (3) exploitation of the extensional fracture set by the downwards migration of Triassic (or later) sandy sediments in an unlithified state. All that can be categorically stated is that the Red Arch Formation is of pre-Permian age. Whilst a Devonian age is preferred, a Lower Carboniferous age cannot be ruled out. Our interpretation of the bedded sedimentary succession concurs with that of Simon (1984*b*).

Possible matches between the palaeomagnetic signature and the apparent polar wander curve allow the sedimentary dykes to be dated as definitely post-Permian possibly late Triassic or even Tertiary. Each of these implied dates have implications for the surrounding and regional geology. The ?ORS succession exposed at Cushendun is overlain to the south and east by undifferentiated Permian–Triassic, Cretaceous Ulster White Limestone and Tertiary Antrim Plateau Basalts. For the sandstone dykes to be late Triassic in age, a simple geological history can be envisaged. Following deposition of the ORS and possibly some overlying Carboniferous, uplift during the Variscan Orogeny brought the ORS to surface. During (ENE–WSW to E–W) extension in the Triassic, red arenaceous sediments typical of the facies developed on the nearby Highland Border Ridge were deposited on the ORS and infilltrated into extensional joints. Both dykes and ORS host then experienced further lithification and burial by the Mesozoic–Tertiary succession listed above. From the Pliocene to the present-day, the area has been subject to erosion of the later cover to a situation at the present-day where the ORS and dykes are at sea level.

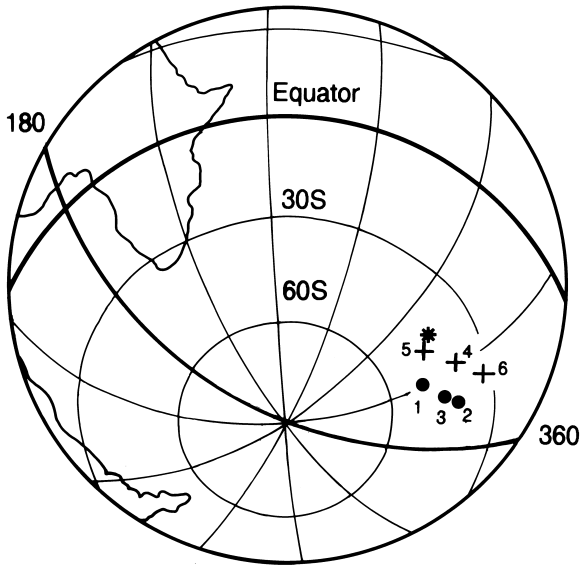


Fig. 11. Palaeomagnetic poles showing the similarity of Late Palaeozoic magnetic overprints in Britain. Poles 1–3 (closed circles) are late Carboniferous/early Permian igneous rocks and poles 4–6 (crosses) are overprints on the Old Red Sandstone. The Star shows the pole for the Red Arch Formation.

Explaining a Tertiary age for the sandstone dykes is also possible, albeit with some added complexity. The geological history of the succession would be largely as that described above except that during Permian–Triassic extension, no joints were filled with arenaceous material. Instead the succession of sediments up to the Tertiary lavas was laid down. After this there are two possible alternative modes of sandstone dyke emplacement. One hypothesis is that early Tertiary extension caused the formation of joints deep enough to penetrate the whole Ulster White Limestone, Permian–Triassic and ORS, which were then filled with a red sandstone unlike any other Tertiary deposit in the vicinity of Cushendun. An alternative explanation for Tertiary-aged dykes is that the area was uplifted in late Cretaceous–early Tertiary times such that the ORS was close to the depositional surface. During this uplift extensional joints were formed which were then infilled with Tertiary sands. Again, no similar sands of this age are recorded nearby, but then this may be a function of later erosion. Open extensional joints thus provided the only accommodation space for the preservation of sediment. This was later lithified, possibly during burial afforded by the Antrim Plateau Basalts, to be re-exhumed in post-basalt times and exposed at the present-day.

Of the two ages (Triassic or Tertiary) and their attendant hypotheses, there are sound geological reasons for a Triassic age to be favoured. The orientation of the sandstone dykes compared to regional measurements of fractures would support a Triassic age (Kerr 1987). Similarly, the Triassic was a time of documented extension in the area (Shelton 1997). Lastly the lithology of the dykes themselves is akin to the overlying arenaceous beds of the Mercia Mudstone Group deposited on and around the Highland Border Ridge. The early Tertiary was a time of uplift and volcanism, albeit with some extension but limited preservation of sediment. No

similar arenaceous Tertiary sediments are known from the area.

To summarize, palaeomagnetic analysis of the sandstone dykes and host rock shows the host to be pre-Permian and the dykes to be post-Permian. On regional geological grounds a late Triassic age is preferred but it is likely that the dykes suffered some remagnetization during Tertiary igneous activity.

Conclusions

(1) The Red Arch Formation is a palaeontologically barren, conglomerate-dominated succession exposed on the Antrim coast of Northern Ireland.

(2) The succession contains sandstone dykes which have a predominant NNW–SSE to WNW–ESE trend.

(3) Palaeomagnetic analyses of the host sandstones and the sandstone dykes shows the presence of two palaeomagnetic components (A and B). The A component has a pole position of 83°N, 346°E and the B component 44°S, 313°E.

(4) The B component is restricted to the host sandstones and is consistent with a late Carboniferous–early Permian age. The exclusively reversed polarity suggests magnetization during the Kiaman Superchron. Since this type of remagnetization is widespread in the Old Red Sandstone, the Antrim section is interpreted as Old Red Sandstone.

(5) The sandstone dykes show no evidence of the Kiaman remagnetization and are therefore considered to be post-Permian in age. The pole position is consistent with a Tertiary age although they may themselves have been remagnetized during this period. The dykes are thought to have been emplaced during a phase of post-Permian extension and the Triassic is considered most likely.

R.S. would like to thank BG plc for financial support. The manuscript was considerably improved thanks to the work of one anonymous referee.

References

Channell, J.E.T., McCabe, C. & Woodcock, N.H. 1992. Early Devonian magnetization directions in Lower Old Red Sandstone of south Wales. *Geophysical Journal International,* **108,** 883–894.

Harris, A.L. 1983. The growth and structure of of Scotland. *In*: Craig, G.Y. (ed.) *Geology of Scotland*. Scottish Academic Press, Edinburgh, 1–22.

Jackson, D.I. & Mulholland, P. 1993. Tectonic and stratigraphic aspects of the East Irish Sea Basin and adjacent areas: contrasts in their post-Carboniferous structural styles. *In*: Parker, J.R. (ed.) *Petroleum geology of Northwest Europe: proceedings of the 4th conference*. The Geological Society, London, 791–808.

Kerr, I.D.V. 1987. *Basement/Cover Structural Relationships in the North Antrim Area Ireland*. PhD Thesis, The Queen's University of Belfast.

McCaffrey, R.J., Buckman, J.O. & Carey, P.F. 1993. Fractured pebbles—A new stress indicator: Comment. *Geology,* **21,** 187.

Mitchell, W.I. & Owens, B. 1990. The geology of the western part of Fintona, Northern Ireland: evolution of Carboniferous basins. *Geological Magazine,* **127,** 407–426.

—— & Turner, N. 1996. A review of the Devonian rocks in Northern Ireland. *In*: *Abstracts of the 39th Annual Irish Geology Research Meeting*. University College, Dublin.

Morris, W.A., Briden, J.C., Piper, J.D.A. & Sallomy, J.T. 1973. Palaeomagnetic studies in the British Caledonides-V. Miscellaneous new data. *Geophysical Journal of the Royal Astronomical Society,* **34,** 69–105.

Sanderson, D.J. 1970. The Highland Border Ridge of North-east Ireland. *Geological Magazine,* **107,** 531–538.

Shelton, R. 1997. Evolution of the Larne Basin, N. Ireland. *In*: Cowan, G. (ed.) *Petroleum Geology of the East Irish Sea & Adjacent areas*. Geological Society, London, Special Publications, **124,** 109–129.

Simon, J.B. 1984*a*. Provenance and depositional history of the Lower Old Red Sandstone of northeast Antrim. *Irish Journal of Earth Sciences,* **6,** 1–13.

—— 1984*b*. Sedimentation of a small complex alluvial fan of possible Upper Old Red Sandstone age. northeast County Antrim. *Irish Journal of Earth Sciences,* **6,** 109–119.

Storetvedt, K.M. & Gidskehaug, A. 1969. The magnetisation of the Great Whin Sill, northern England. *Physics of the Earth and Planetary Interiors,* **2,** 105–114.

Symes, R.G. & McHenry, A. 1886. *Explanatory memoir to accompany sheet 14*. Memoirs of the Geological Survey of Ireland.

Torsvik, T.H. 1985. Paleomagnetic results from the Peterhead granite, Scotland; Implication for regional late Caledonian magnetic overprinting. *Physics of the Earth and Planetary Interiors,* **39,** 108–117.

Turner, P. 1980. *Continental Red Beds*. Elsevier. Developments in Sedimentology **29**.

—— & Archer, R.A. 1975. Magnetisation history. of Lower Old Red Sandstones from the Gamrie Outlier, Scotland. *Earth and Planetary Science Letters,* **27,** 240–250.

Wilson, H.E. 1953. The petrography of the Old Red Sandstone Rocks of the north of Ireland. *Proceedings of the Royal Irish Academy,* **55B,** 83–320.

Zijderveld, J.D.A. 1967. The natural remanent magnetization of the Exeter volcanic traps (Permian, Europe). *Tectonophysics,* **4,** 121.

Ziegler, P. 1982. *Geological Atlas of Western and Central Europe*. Elsevier, Amsterdam.

Received 19 April 1998; revised typescript accepted 2 June 1999.
Scientific editing by Peter Haughton.